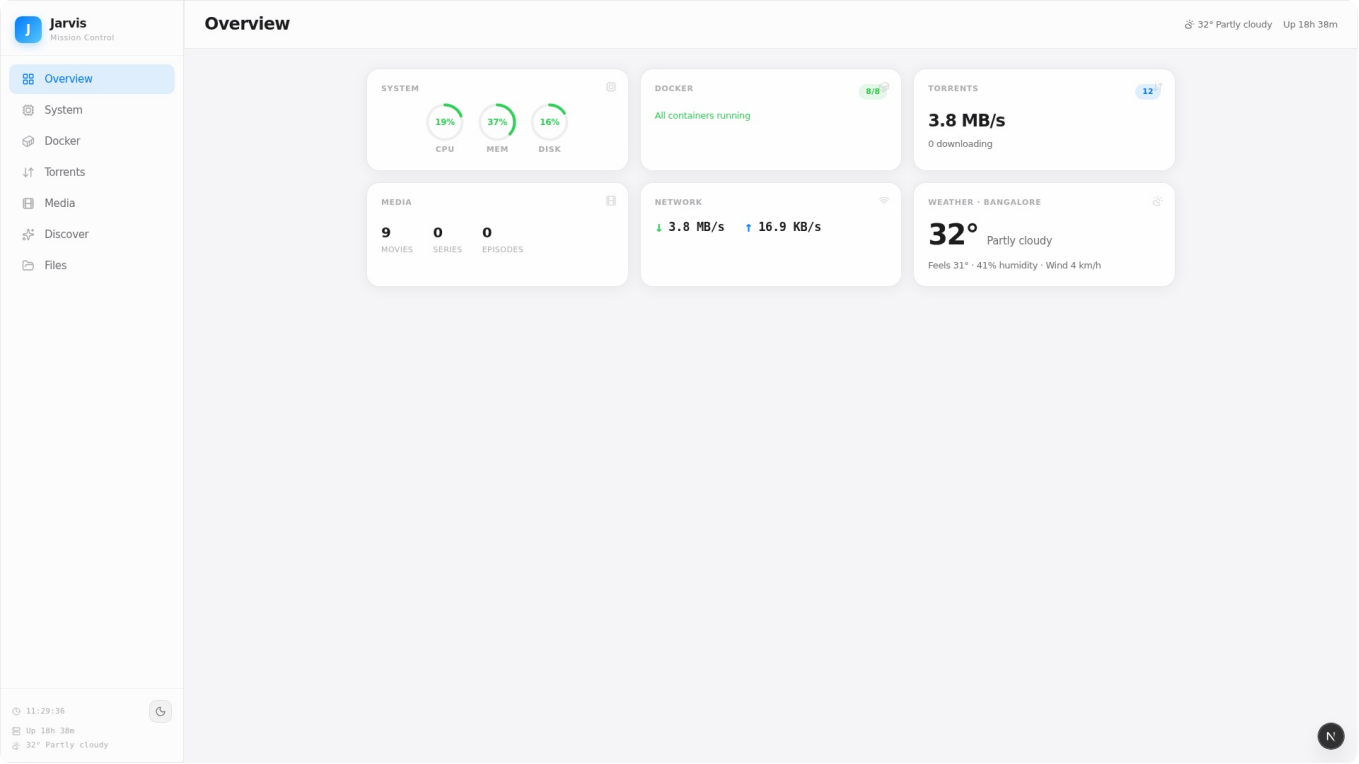
Jarvis Dashboard -- Homelab Mission Control

A single-pane-of-glass dashboard for managing your entire homelab: system monitoring, Docker, torrents, media, file management, and an AI-powered movie recommendation engine.



What is this?

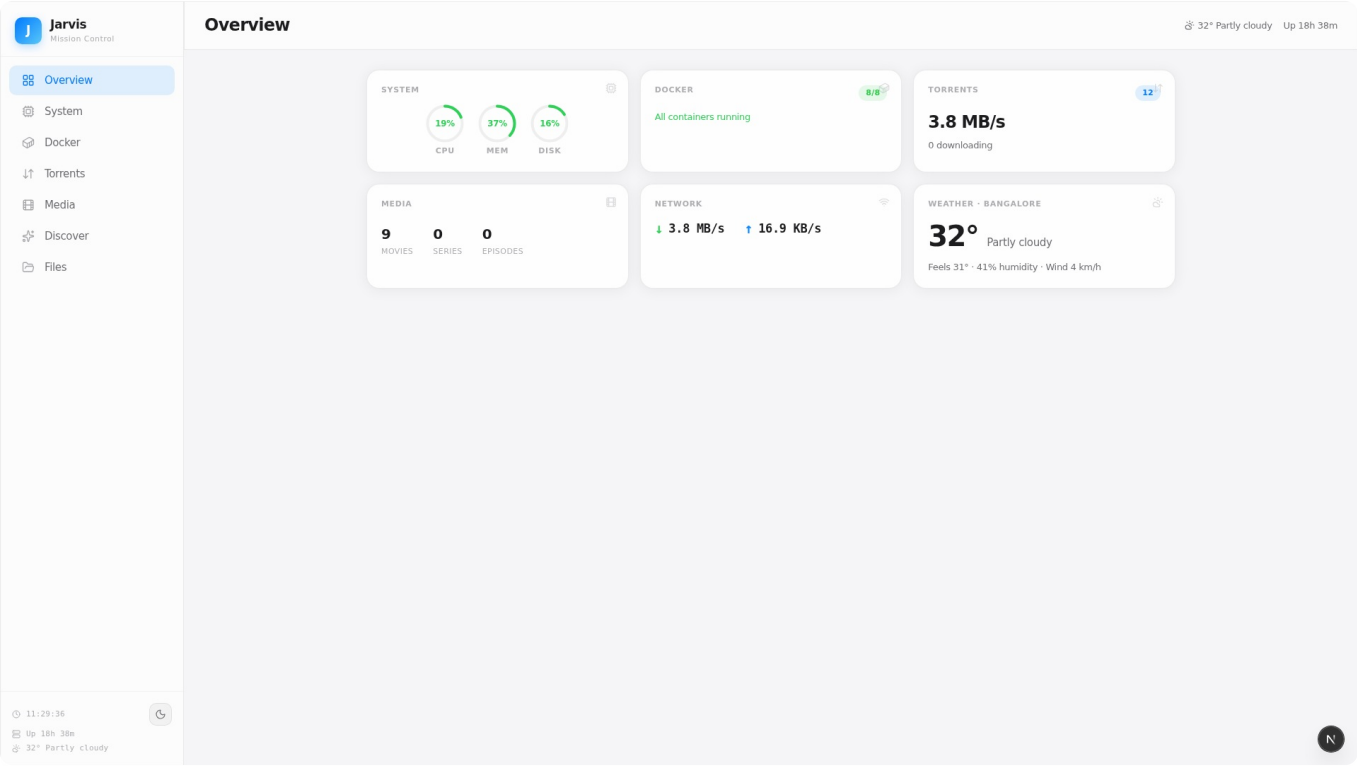
Jarvis is a self-hosted homelab dashboard that replaces the need to SSH into your server or juggle a dozen browser tabs. It brings together real-time system stats, Docker container management, torrent operations, Jellyfin media tracking, a full file explorer, and a movie recommendation engine powered by Reddit community data and TMDB -- all wrapped in a clean, responsive interface with dark and light modes.

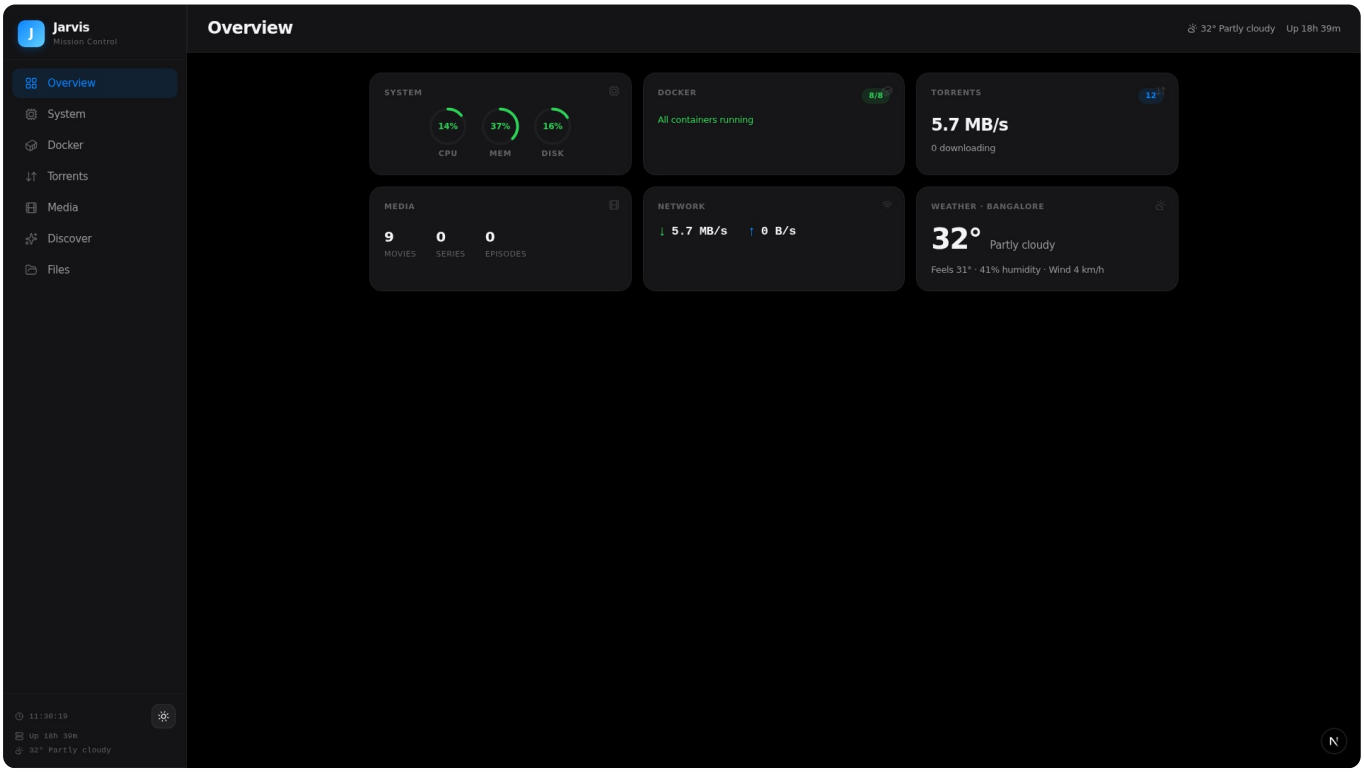
Features

Feature	Description
System Monitoring	Real-time CPU, RAM, disk gauges with bandwidth history graph and top processes
Docker Management	Container grid with status indicators, CPU/memory bars, start/stop/restart, live log viewer
Torrent Integration	Search, add, and manage torrents via qBittorrent with progress tracking
Jellyfin Media	Library stats, recently added, now playing sessions
Movie Recommendations	Mood-based, similar-to, library analysis, and trending -- powered by Reddit + TMDB
TMDB Movie Search	Search any movie or series with full detail pages (cast, ratings, posters, synopsis)
File Explorer	Browse, rename, copy, move, delete, download -- with breadcrumbs and context menus
Dark / Light Mode	System-aware theme toggle with CSS variables
Responsive UI	Desktop sidebar + mobile bottom navigation

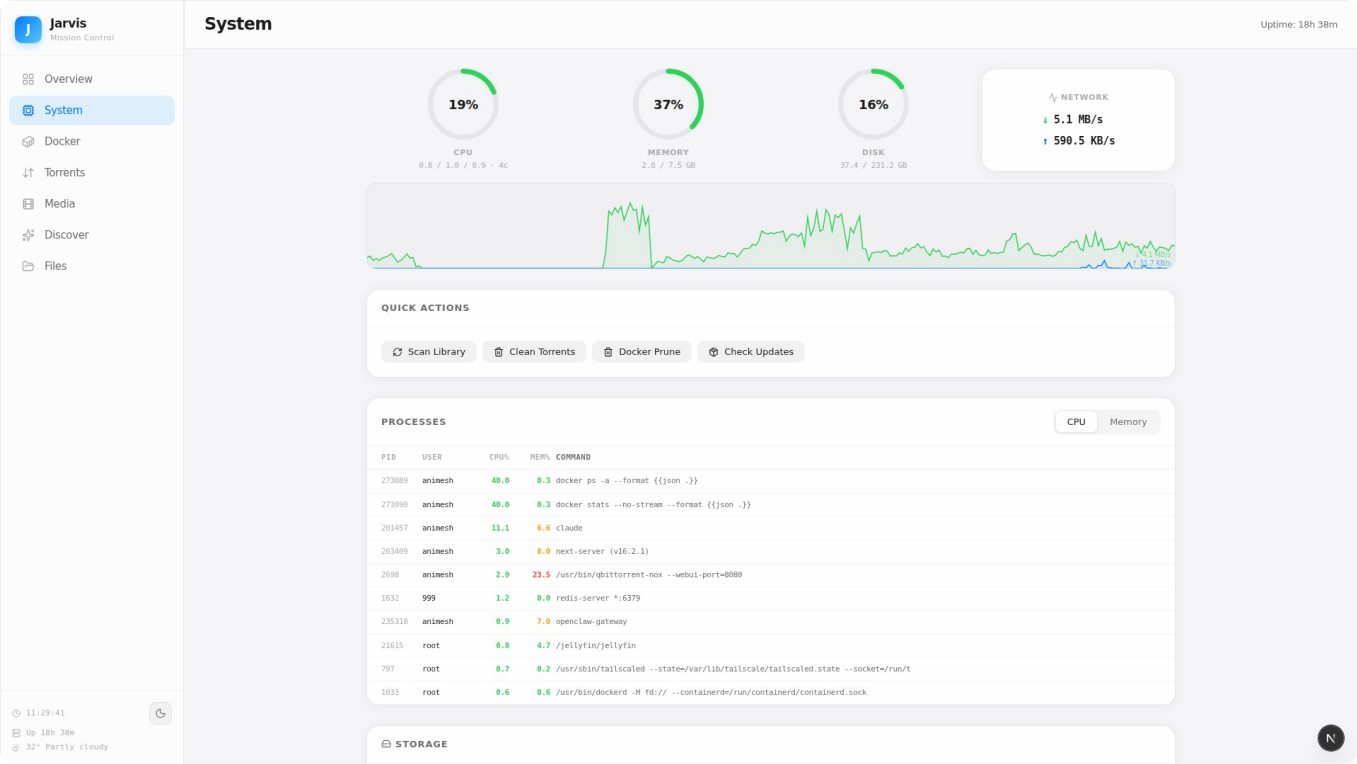
Screenshots

Overview Dashboard

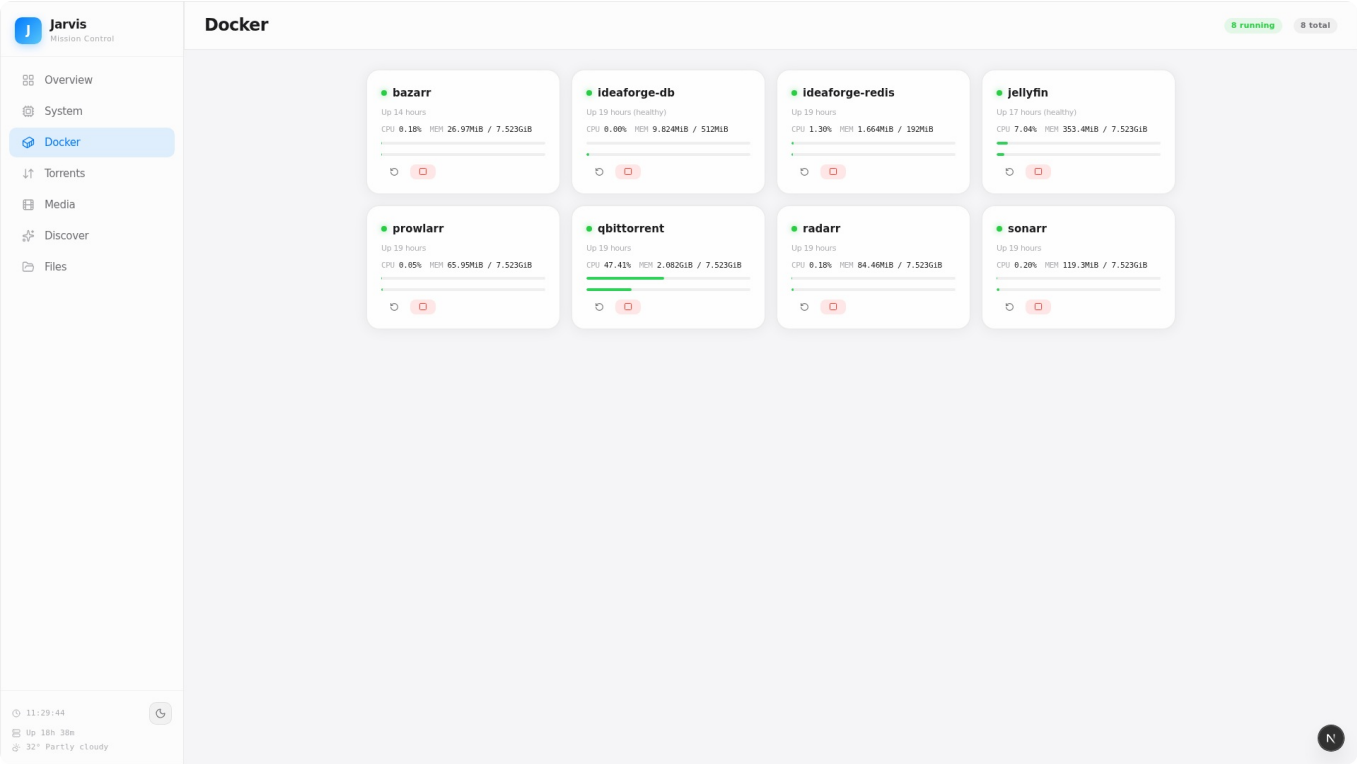




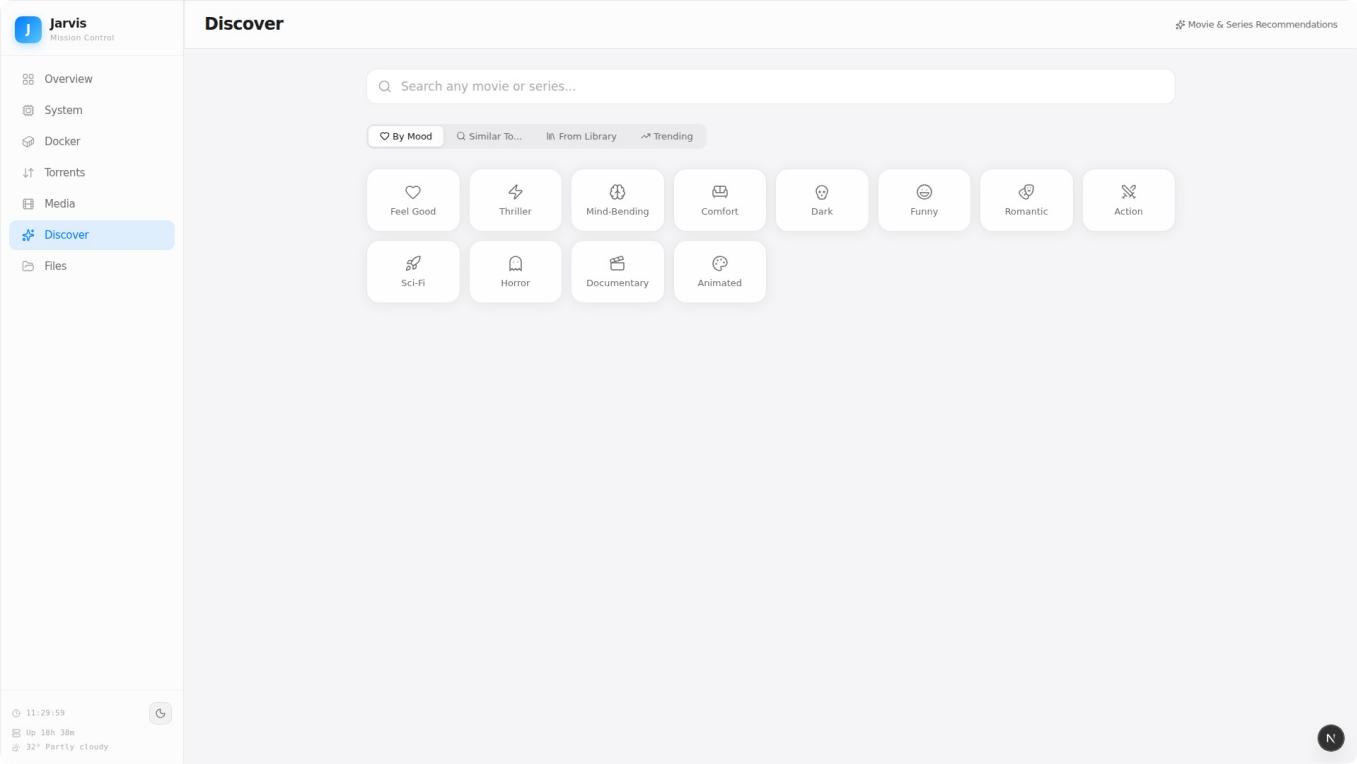
System Monitoring



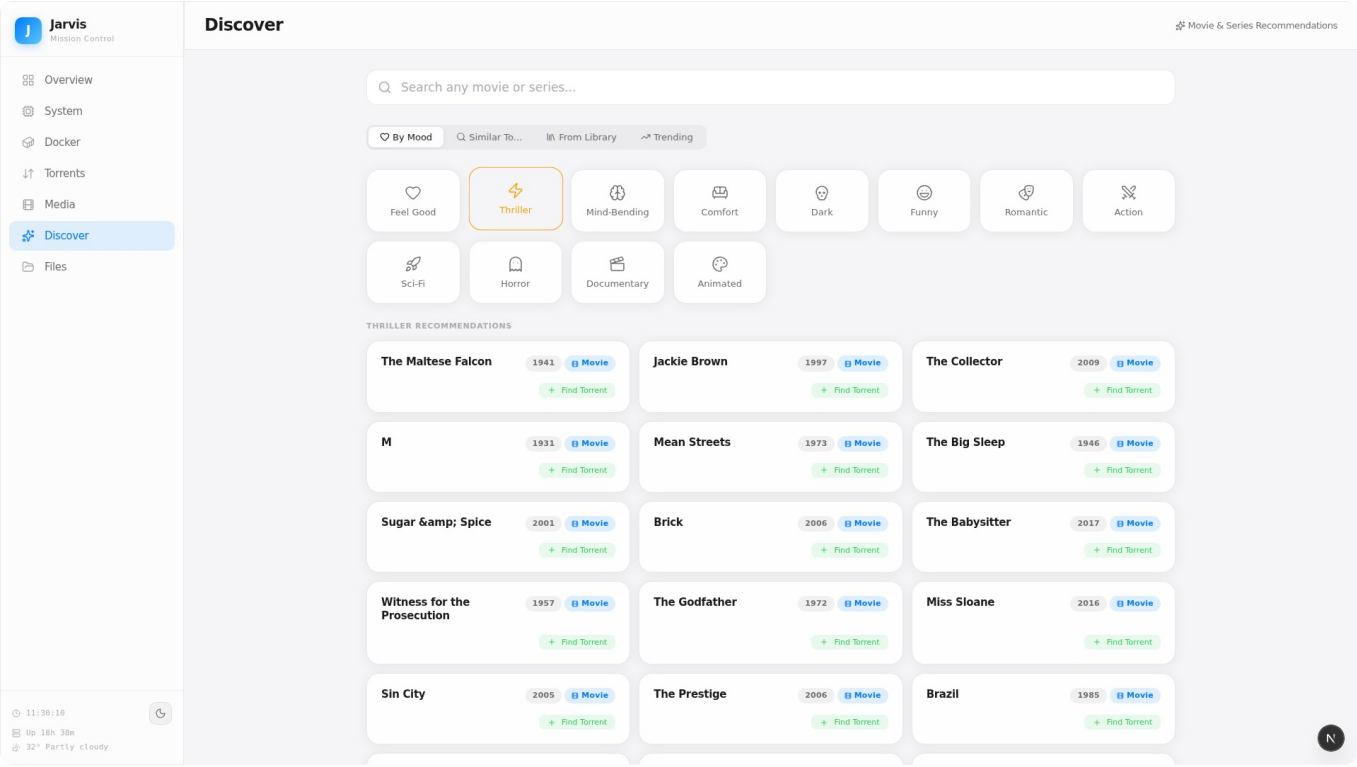
Docker Management



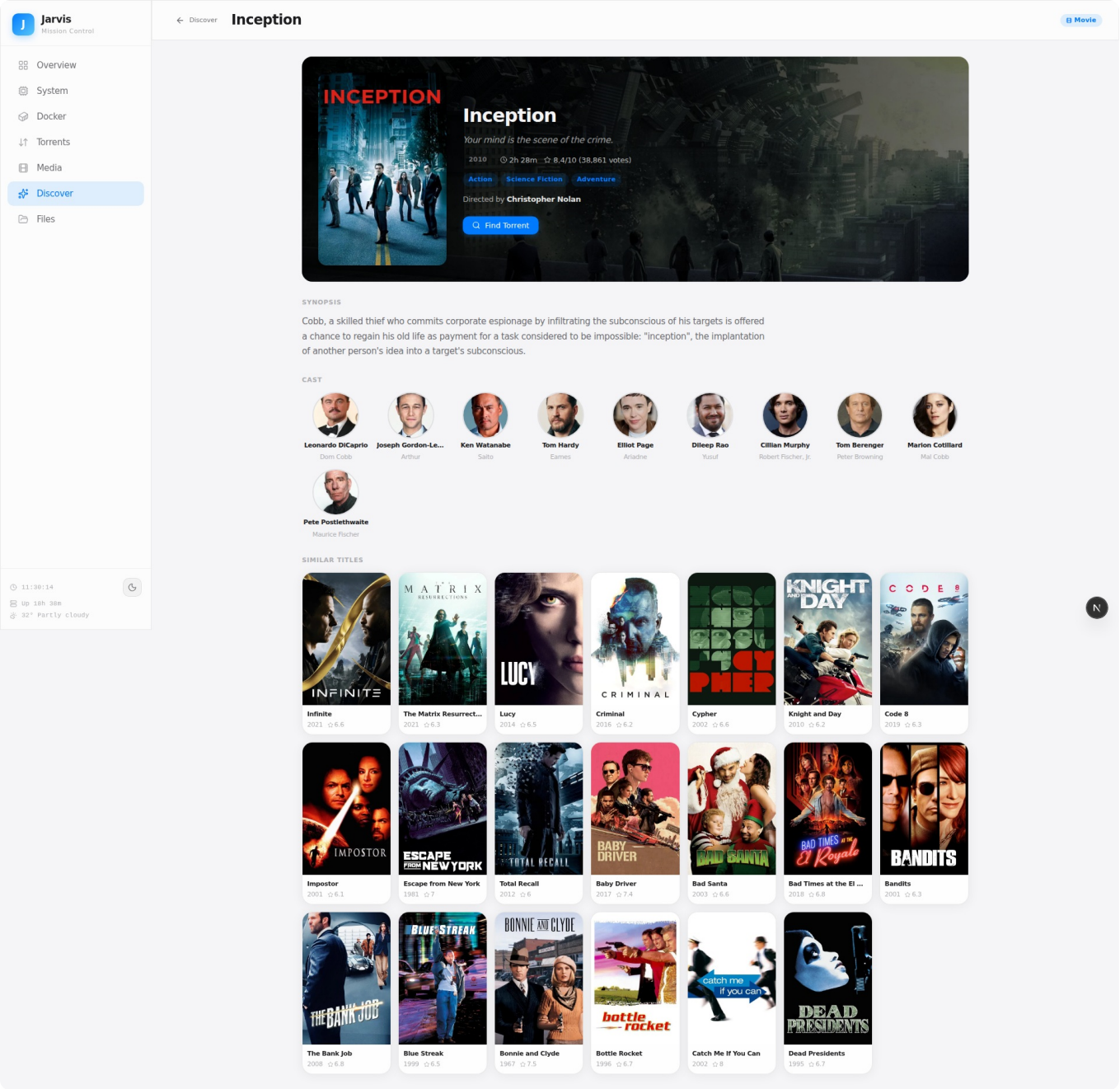
Movie Discovery -- Mood Picker



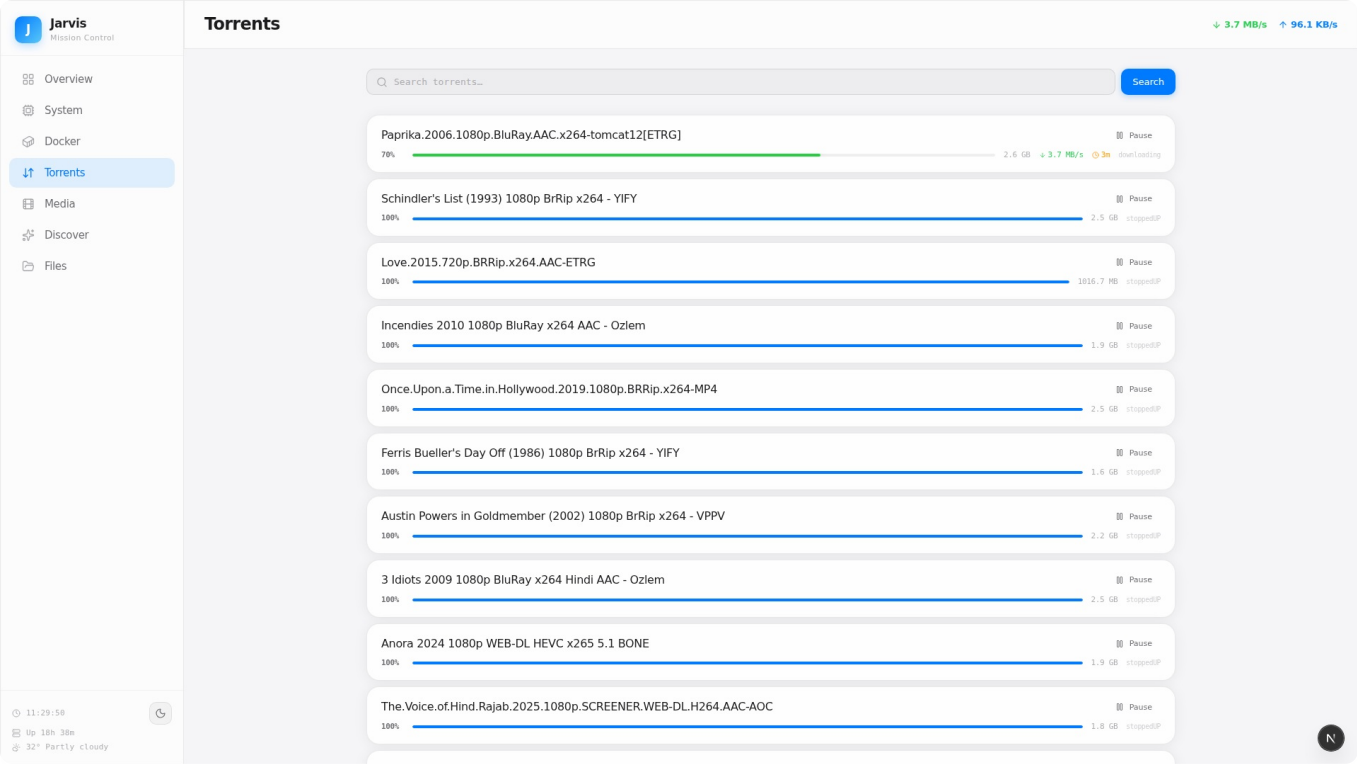
Movie Discovery -- Results



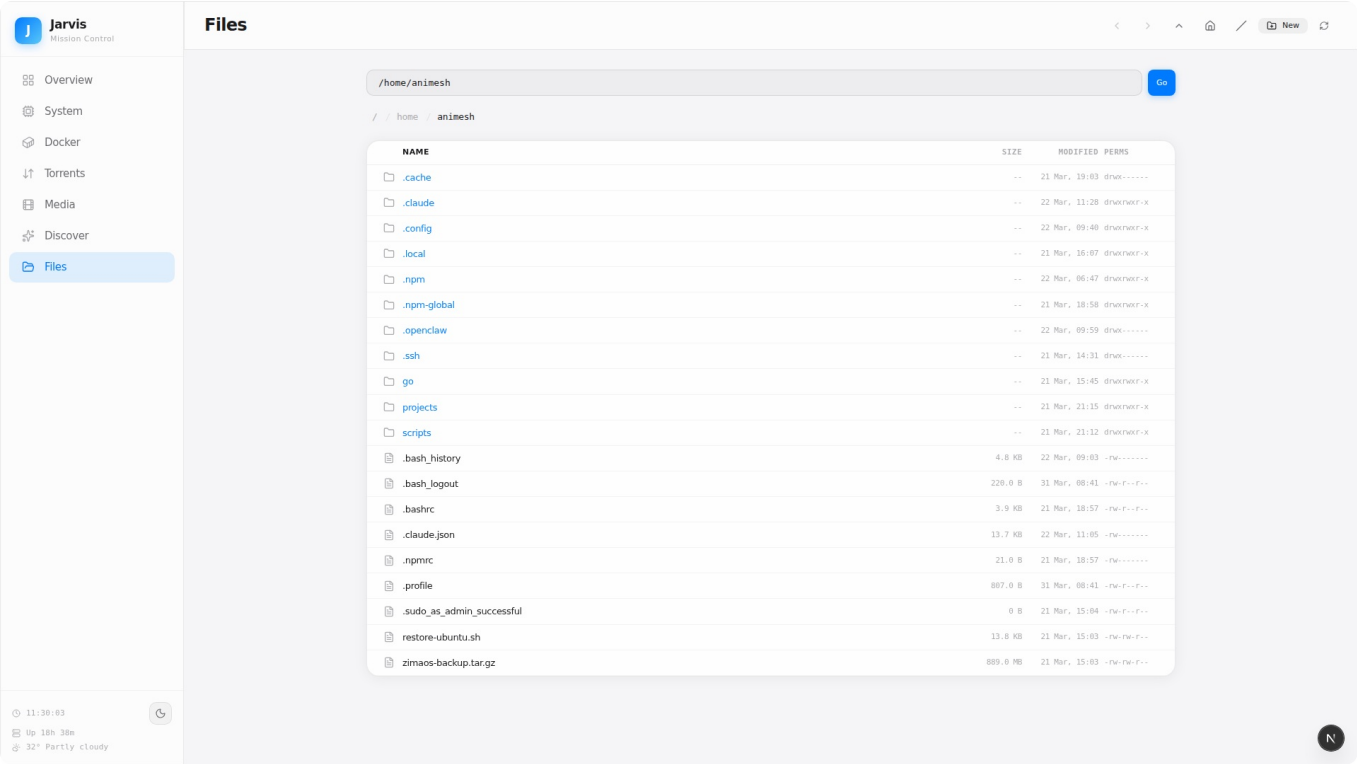
Movie Detail Page



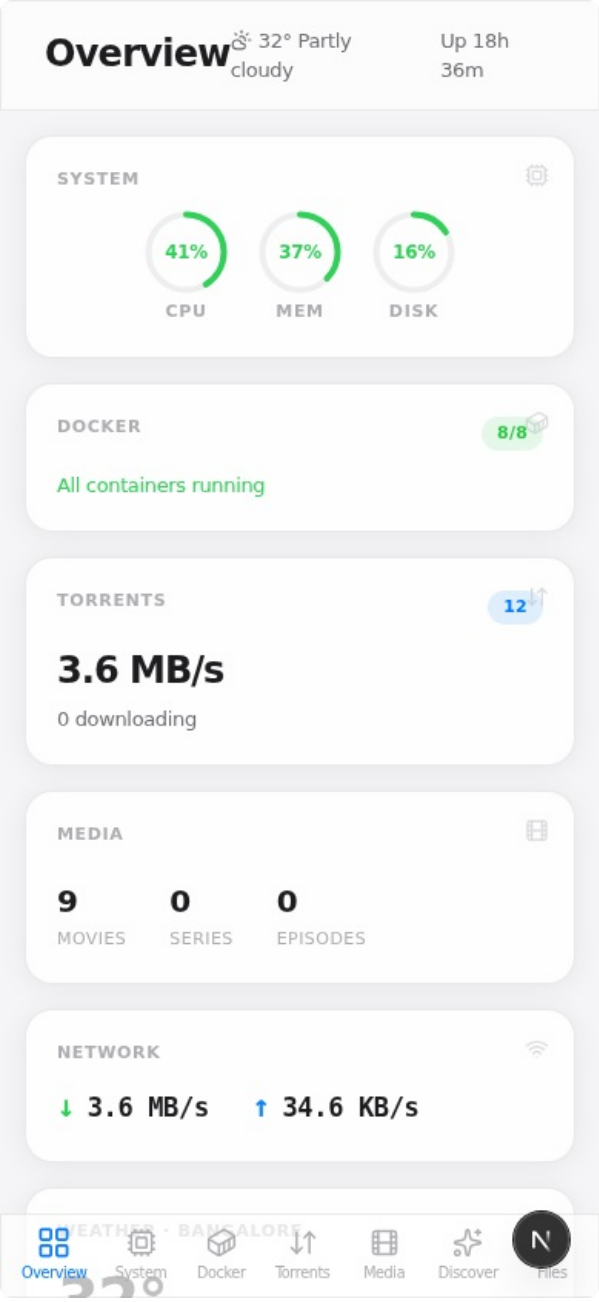
Torrent Management



File Explorer



Mobile View



Architecture

```
Browser --> Next.js (3000) --> Python Backend (8002) --> Docker / Jellyfin / qBittorrent / TMDB / Reddit
```

Layer	Tech
Frontend	Next.js 16, React 19, TypeScript, SCSS Modules
Backend	Python 3 (stdlib only -- no Flask/Django/pip installs)
Icons	Lucide React
Theme	CSS Variables + localStorage
Data Sources	Jellyfin API, qBittorrent API, Docker CLI, TMDB API, Reddit Wiki, wttr.in

The backend is a single `server.py` file using Python's built-in `ThreadingHTTPServer`. It acts as an API proxy (avoiding CORS issues), runs system/Docker commands via subprocess, and serves as the bridge between the Next.js frontend and all

external services. Zero dependencies beyond the Python standard library.

Recommendation Engine

This is the standout feature. The movie recommendation system combines curated community data with live API enrichment:

Data source: Reddit r/MovieSuggestions wiki -- 2,896 curated movies across 120 categories, scraped and indexed locally.

Four discovery modes:

Mode	How it works
By Mood	Pick from 12 moods (Feel Good, Thriller, Mind-Bending, Horror, etc.) to get curated picks
Similar To	Type a movie name with autocomplete, get community-recommended similar titles
From Library	Analyzes your Jellyfin library genres to suggest titles you don't already own
Trending	Current trending movies and series from TMDB

Every recommendation links to a full detail page with TMDB data (cast, ratings, poster, synopsis), and a one-click "Find Torrent" button that searches and adds torrents directly from the recommendation card.

Setup

```
# 1. Clone the repo
git clone https://github.com/your-username/jarvis-dashboard.git
cd jarvis-dashboard

# 2. Configure environment
cp .env.example .env
# Edit .env with your credentials:
#   JELLYFIN_API_KEY=your_key
#   QBIT_USER=admin
#   QBIT_PASS=your_password
#   TMDB_API_KEY=your_tmdb_key

# 3. Start the backend
python3 server.py &

# 4. Start the frontend
cd frontend
npm install
npm run dev

# 5. Open http://localhost:3000
```

Requirements: Python 3.8+, Node.js 18+, Docker (for container management features), Jellyfin and qBittorrent running on the same network.

▼ API Endpoints

System

Method	Endpoint	Description
GET	`/api/system`	CPU, memory, disk, uptime
GET	`/api/processes`	Top 10 processes by CPU and memory
GET	`/api/storage`	Media directory sizes (cached 5min)

GET	`/api/weather`	Weather via wttr.in (cached 15min)
GET	`/api/bandwidth/history`	Network speed history (last 30min)

Docker

Method	Endpoint	Description
GET	`/api/docker/containers`	List all containers with status
GET	`/api/docker/stats`	Container resource usage
GET	`/api/docker/logs?container=X&lines=N`	Container log output
POST	`/api/docker/action`	Start, stop, or restart a container

Torrents

Method	Endpoint	Description
GET	`/api/torrent-search?q=X`	Search torrents via apibay
POST	`/api/torrent-add`	Add magnet link to qBittorrent
GET/POST	`/api/qbit/*`	Proxy to qBittorrent Web API

Media

Method	Endpoint	Description
GET	`/api/jellyfin/*`	Proxy to Jellyfin API (items, sessions, system info)

Recommendations

Method	Endpoint	Description
GET	`/api/recommendations/mood?mood=X`	Get recommendations by mood
GET	`/api/recommendations/similar?title=X`	Find similar titles
GET	`/api/recommendations/library`	Recommendations based on Jellyfin library
GET	`/api/recommendations/trending`	Currently trending on TMDB
GET	`/api/recommendations/categories`	List all available categories
GET	`/api/recommendations/autocomplete?q=X`	Title autocomplete for search
GET	`/api/recommendations/search?q=X`	Full TMDB search
GET	`/api/recommendations/detail?type=X&id=Y`	Full movie/series detail from TMDB

Files

Method	Endpoint	Description
GET	`/api/files/list?path=X`	List directory contents
GET	`/api/files/download?path=X`	Download a file

POST	`/api/files/delete`	Delete file or directory
POST	`/api/files/move`	Move file or directory
POST	`/api/files/copy`	Copy file or directory
POST	`/api/files/mkdir`	Create new directory
POST	`/api/files/rename`	Rename file or directory

Quick Actions

Method	Endpoint	Description
POST	`/api/actions/jellyfin-scan`	Trigger Jellyfin library scan
POST	`/api/actions/clean-torrents`	Remove completed torrents
POST	`/api/actions/docker-prune`	Prune unused Docker resources
POST	`/api/actions/update-check`	Check for system updates

Credits

- Built with Claude Code -- AI pair programming
- Movie data: TMDB, Reddit r/MovieSuggestions community
- Icons: Lucide
- Weather: wttr.in

License

MIT